MATH290

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Proposition 1. *For any integer k, $k^3 - k$ is a multiple of 6.*

Proof.

Firstly, recall that an integer is a multiple of 6 if it can be expressed as the product of 6 and some other integer. Now, observe what happens if you substitute $-k$ for k:

$$\begin{aligned}&(-k)^3 - (-k)\\ =&-(k^3) + k\\ =&-(k^3 - k)\end{aligned}$$

Thus, if $k^3 - k$ can be expressed as $6n$ for some integers k and n, then $(-k)^3 - (-k)$ can be expressed as $-6n$. Therefore, if I can show that $k^3 - k$ is a multiple of 6 for all integers $k \geq 0$, then $k^3 - k$ is a multiple of 6 for all integers $k < 0$. With this in mind, I'll only focus on integers $k \geq 0$ for the rest of the proof.

Let f_k be the sequence $k^3 - k$ for integers $k \geq 0$. It turns out that rather than proving each term in this sequence is a multiple of 6, it's easier to prove that *the difference between consecutive terms* is a multiple of 6. (Or at least *I* think it is.) Let's let g_k be the difference between any f_{k+1} and f_k. That is,

$$\begin{aligned}g_k &= f_{k+1} - f_k\\ f_{k+1} &= f_k + g_k\end{aligned}$$

If I can show that g_k is a multiple of 6 for all integers k, then so long as the sequence starts with a multiple of 6 all following terms will be multiples of 6. This relies on the property that the sum of two multiples of 6 is itself a multiple of 6. I'll briefly demonstrate this property for two multiples of six $n = 6i$ and $m = 6j$, where n, m, i, and j are all integers. Observe,

$$\begin{aligned}n + m &= 6i + 6j\\ &= 6(i + j)\end{aligned}$$

Relying on the property that summing integers yields an integer, we've shown that $n + m$ can be expressed as a product of 6 and some other integer $i + j$, and thus $n + m$ is a multiple of 6.

□